



Federal Board SSC-II Examination

Physics Model Question Paper

(National Curriculum of Pakistan 2022-2023)
(Scheme of Studies 2006)

Section - A (Marks 12)

Time Allowed: 20 minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

ROLL NUMBER					
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(1)	(1)	(1)	(1)	(1)	(1)
(2)	(2)	(2)	(2)	(2)	(2)
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(4)	(4)	(4)	(4)	(4)	(4)
(5)	(5)	(5)	(5)	(5)	(5)
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Candidate Sign. _____

Invigilator Sign. _____

Q1. Fill the relevant bubble against each question according to curriculum. Each part carries one mark.

Sr #	Question	(A)	(B)	(C)	(D)	(A)	(B)	(C)	(D)
1.	When light waves pass from air to glass and undergo refraction, which property does remain constant?	Direction	Speed	Frequency	Wavelength	☐	☐	☐	☐
2.	A student rubs a plastic rod with a cloth. The rod becomes positively charged. What has happened to the rod?	It has gained electron	It has gained proton	It has lost proton	It has lost electron	☐	☐	☐	☐
3.	Which group of two logic gates has HIGH output when both of their inputs are HIGH?	AND and OR	AND and NOR	NAND and NOR	NAND and OR	☐	☐	☐	☐
4.	A girl warms her hands by holding them near a fire. Using heat transfer principles, tell how the heat reaches her hands.	Conduction only	Conduction and convection	Radiation only	Convection and radiation	☐	☐	☐	☐
5.	Force on a current carrying conductor will be maximum if angle between rod and magnetic field is:	90°	45°	0°	180°	☐	☐	☐	☐
6.	A current of 1 ampere is drawn by a filament of bulb. Number of electrons passing through a cross-section of the filament in 16 seconds would be:	10 ²⁰	10 ¹⁹	10 ⁻²⁰	10 ⁻¹⁹	☐	☐	☐	☐

7.	Relation between coefficient of linear and volume expansion is:	$\alpha = 3\beta$	$\alpha = \beta/3$	$\beta = \alpha/3$	$\beta = \alpha$	☐	☐	☐	☐
8.	Which of the following has more ionization power?	Alpha-particle	Beta - particle	Gamma ray	Neutrons	☐	☐	☐	☐
9.	Which of the following characteristics of a wave is independent of others?	Speed	Frequency	Amplitude	Wavelength	☐	☐	☐	☐
10.	In a DC motor, commutators are in motion. Which of the following remains unaffected by their movement?	Current	Direction of forces	Direction of torque	Position of commutators with carbon brushes	☐	☐	☐	☐
11.	Sound travels faster in:	Plastic	Steel	Air	Oil	☐	☐	☐	☐
12.	What happens to an ordinary silicon diode when it is forward biased?	It conducts electricity	It blocks electricity	It emits light	It stores energy	☐	☐	☐	☐



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Time allowed: 2.40-hour

Total Marks Sections B and C: 53

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the **E-sheet**. Write your answers on the allotted/given spaces.

SECTION – B (Marks 33)

Q.2 Answers the following questions briefly.

(11x3 = 33)

Q. 2	Questions	Marks	Questions		Marks
(i)	What is meant by isotope of an element? Give examples of three isotopes of any element.	03	OR	What is meant by specific heat capacity of a substance? Write its formula and SI unit.	03
(ii)	Define electric field. How is it expressed mathematically? Is it a vector quantity?	03	OR	What is meant by biasing of a diode? What are types of biasing?	03
(iii)	How do longitudinal waves differ from transverse waves? Give some real-life examples of longitudinal and transverse waves.	03	OR	Two identical batteries, each with an emf of 1.5V, are connected (a) in series and (b) in parallel. Calculate the total emf in each case.	03
(iv)	What is the cost of running four light bulbs of 100 watts each for 6 hours in a school for a month? Cost of electricity unit is Rs. 12 per kWh.	03	OR	Calculate the speed of light in a medium of refractive index 1.5.	03
(v)	With the help of an example, show that Lenz's law follows conservation of energy.	03	OR	How are electrical energy and electrical power related? How can you calculate each one in a simple electric circuit?	03
(vi)	Elaborate the differences between alternating current and direct current by using current-time graphs.	03	OR	How do land and sea breezes form, and what role do temperature difference between land and sea water play in these breezes?	03
(vii)	What is an echo? Calculate the minimum distance between source and obstacle to hear echo in air? (speed of sound in air = 340 m/s).	03	OR	What are the common methods used to discharge an insulator? Briefly elaborate any one method.	03
(viii)	What is the electromagnetic spectrum? Give its order based on frequency of radiation.	03	OR	Define ohm's law. What are its limitations?	03
(ix)	A radioactive substance has a half-life of 6 hours. Its initial amount is 300 grams. Calculate the amount left after 18 hours.	03	OR	A convex lens with a focal length of 10 cm produces a linear magnification of 3. A 5 cm tall object is placed 30 cm in front of it. Calculate the image distance and the height of the image formed.	03
(x)	When will a converging lens act as a magnifying glass? Show it by a ray diagram.	03	OR	How does the wavelength of a wave change if its frequency is doubled in a medium while its speed remains constant?	03
(xi)	Differentiate between evaporation and boiling of liquids.	03	OR	Why does the sky appear blue during the day?	03

SECTION – C (Marks 20)

Note: Attempt all questions. Marks of each question are given.

Q.3	What is conduction? Explain it on the basis of kinetic theory of solids. What makes metals better conductors than other substances?	05	OR	What is nuclear fusion reaction? Write down steps involved in proton- proton cycle. What is cause of energy in nuclear fusion?	05
Q.4	Draw a comparative sketch illustrating the loudness and pitch of sound waves. Also give their brief description.	05	OR	Explain the phenomenon of total internal reflection. What are necessary conditions of total internal reflection? Derive formula for refractive index of a medium in term of critical angle.	05
Q.5	A solid cube of side length 5cm at 15°C is heated. Calculate its (a) initial volume (b) increase in its volume (c) final volume at 100°C . (Coefficient of volume expansion of this solid is $9 \times 10^{-6} \text{ K}^{-1}$).	05	OR	Three resistors 5Ω , 10Ω and 15Ω are connected in parallel combination across 30V battery. Calculate the equivalent resistance, total current and current across each resistor?	05
Q.6	What is meant by transformer? Explain the working of a simple iron-cored transformer. Give transformer equation.	05	OR	Differentiate between digital and analogue electronics. What makes digital system better than an analogue system?	05